

OVERTONE

Technical briefing — meeting join & live grounding

Architecture Overview

Audience: Technical partnership discussion
Status: Active development — demo / architecture stage
Date: August 2026
Classification: Confidential — partner review

1. Framing

Overtone is a **live meeting layer**: a bot joins Google Meet / Zoom / Teams as a participant, presents a deck, listens with realtime speech-to-speech, and answers from **indexed session sources** — with a design goal to **abstain or escalate** when grounding is weak (not invent).

It is **not** a finished “zero-hallucination guarantee” product. The honest framing is: **traceable, source-bound outputs + hard stop when the source cannot support an answer.**

2. System context

Operator → Dashboard (upload & launch) → Backend (FastAPI) → Recall.ai bot → Meeting room. Recall opens the Presenter UI as the bot camera. Presenter ↔ Backend over WebSocket (audio + tools). Backend ↔ Realtime voice (Gemini Live preferred / OpenAI Realtime). Backend reads slide index (pgvector), GCS artifacts, and Postgres session catalog.

Component	Role
Dashboard	Upload decks, configure agents, launch bots into meetings
Recall.ai	Meeting participant join; virtual camera / audio path
Presenter UI	Deck display as bot camera; mic capture; play spoken audio
Backend relay	Session state, tool execution, voice bridge
Realtime voice	Speech-to-speech + function / tool calling

Component	Role
pgvector + Vision	Per-page extract, embed, retrieve grounded slide text
GCS + Postgres	Artifacts and durable session catalog

3. Session lifecycle (summary)

- Upload PDF / PPTX → store → Vision extract per page → quality gate → embeddings → pgvector → status ready
- Launch: meeting URL + knowledge base → create session → Recall bot with output-media presenter URL
- Bot joins room → presenter WebSocket → Live voice session
- Participant audio → relay → model; model tool calls → navigate / get_slide_details / search → spoken answer from tool results only

4. Platform layers

Layer	Function	Current technology
Capture / join	Enter call as participant; camera shows slides	Recall.ai
Presenter	Deck UI, mic, bot audio playback	React (output-media page)
Realtime voice	Speech-to-speech + tools	Gemini Live / OpenAI Realtime
Orchestration	State, tools, mute, page position	FastAPI relay + executor
Grounding	Per-slide retrieval	Vision → embeddings → pgvector
Persistence	Files, catalog, session extras	GCS + Postgres

5. Grounding path

User question → tool routing (navigate / get_slide_details / search) → indexed slide text + session page. Enough evidence → spoken answer. Insufficient evidence → abstain / clarify → escalate path (roadmap).

Design principles (escalate-don't-guess)

- Prefer the **currently visible slide** for “explain this / what's on this.”
- Search the **deck index** before any external lookup.
- If retrieval is empty or stub-thin → **do not invent**; clarify or state not in source.
- **Human escalate** (expert as ground truth) is the intended next architectural step — not claimed as fully shipped today.

6. Indexing pipeline

Upload → store (GCS / local) → render page images → Vision per page → title / body / searchable_content → quality gate (reject stub pages) → embeddings → pgvector → presentation status ready. Vision stubs such as title-only “Slide 4” must not be marked ready.

7. Realtime tools

Tool	Role
navigate_to_slide	Jump to a page; persist current_page
get_slide_details	Narrate current or specified page from index
search_and_answer	Hybrid retrieve over slides; stay or navigate
mute_self / unmute_self	Hard silence path
leave_call	Explicit exit only
fetch_external_data	Last resort; not for on-deck questions

Spoken answers are instructed to come from **tool results** (slide_content / brief_content), not model memory.

8. Maturity

Solid enough to demonstrate	Explicit next (roadmap)
Join Meet via Recall; present deck as bot camera; realtime voice; tool-grounded slide Q&A; index quality gate	First-class escalate-to-human queue; stronger abstention at tool boundary; multi-instance WS fan-out; claim→source audit trail

9. Fit with a digital protégé product

Protégé concern	Overtone role
Join Meet / Zoom as participant	Meeting adapter (focus of current demos)
Follow conversation without guessing	Session + deck grounding + abstain
Respond when addressed	Realtime voice + tools
Escalate when unsure	Architectural fit — align with expert KB / human handoff

Overtone is best understood as a **live meeting adapter + grounding shell**, not a replacement for an expert-judgment knowledge system.

This document describes current engineering architecture as of the date on the cover. Capabilities labeled roadmap are intentional design direction, not production guarantees.

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